

Subject Description Form

Subject Code	ABCT5108
Subject Title	Investment, Financing and Risk Management in BioBusiness
Credit Value	3
Level	5
Pre-requisite	Nil
Objectives	1. Industry Insight: Achieve a thorough understanding of the biotechnology and life sciences sector, encompassing its trends, challenges, and opportunities. 2. Financial Management: Attain expertise in evaluating and overseeing the financial operations within the biotechnology business landscape. 3. Business Development: Master the art of spotting and executing strategic business development opportunities in the biotech industry. 4. Risk Management: Equip students with the ability to discern, evaluate, and mitigate various risks associated with biotech enterprises.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) acquire comprehensive knowledge about the biotechnology industry, including its structure, trends, challenges, and opportunities. b) develop skills in analyzing financial statements, managing budgets, and applying valuation methods in biobusiness entities. c) understand various funding sources for biotech companies, including venture capital, private equity, public funding, and strategic partnerships. d) learn and apply strategies related to market analysis, technology transfer, strategic alliances, and intellectual property management in biobusiness. e) gain the ability to identify, assess, and manage various risks, including regulatory, clinical, and operational, in the biotechnology sector.
Subject Synopsis/ Indicative Syllabus	Module 1: Introduction to Biobusiness (3 hours) • Overview of the biotechnology and life sciences industry • Trends, challenges, and opportunities in the biotech sector • Role of finance, business development, and risk management in biobusiness Module 2: Financial Principles in Biobusiness (6 hours) • Financial statements analysis • Financial ratios and metrics • Valuation methods for biotech startups and established companies • Budgeting and financial forecasting Module 3: Funding and Financing in Biotechnology (6 hours) • Sources of funding for biotech startups and established companies • Venture capital and private equity investments in the biotech sector

	- Public funding options, including government grants and subsidies - Debt financing and strategic partnerships in biobusiness Module 4: Business Development Strategies (9 hours) - Market analysis and opportunity assessment in the biotechnology industry - Licensing and technology transfer in biobusiness - Strategic alliances, mergers, and acquisitions in biotech - Intellectual property protection and commercialization strategies Module 5: Risk Management in Biobusiness (9 hours) - Identification and assessment of risks in the biotech sector - Regulatory and compliance risks in biotechnology - Clinical trials and drug development risks - Risk mitigation strategies and insurance in biobusiness Module 6: Case Studies and Practical Exercises (6 hours) - Analysis of real-world biotech case studies - Group discussions and presentations on selected topics - Practical exercises and simulations to apply concepts learned																															
Teaching/Learning Methodology	Lectures, Assignments, and Examination Final Presentation - Facilitates a nuanced understanding of the field. Collaborative group work and peer review processes will be integral to the course, promoting teamwork and critical analysis. Interactive seminar sessions, including lively discussions and Q&A segments, will foster active engagement and reinforce key learning objectives. Presentations will be evaluated based on their depth of research, clarity of communication, and the ability to provoke thoughtful discussion. Written Assignments - A written assignment where students delve deep into topics like the role of insurance in BioBusiness, taxation implications, and the regulatory impact on investment decisions. In-class participation - Students are expected to attend the classes to generate insightful discussion.																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">Percentage weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>1. Group presentation</td><td>50%</td><td>✓</td><td></td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>2. In-class participation</td><td>10%</td><td></td><td>✓</td><td></td><td>✓</td><td>✓</td></tr> </tbody> </table>						Specific assessment methods/tasks	Percentage weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Group presentation	50%	✓		✓	✓		2. In-class participation	10%		✓		✓	✓
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1. Group presentation	50%	✓		✓	✓																											
2. In-class participation	10%		✓		✓	✓																										

	3. Midterm Quiz	40%	✓	✓	✓	✓	✓
	Total	100%					
Students are allowed to use GenAI tools to support their writing of and essays. If GenAI tools are used to support their essay writings, students must declare the use of such tools and how they have been used in the assessments. It should be noted that submitting a work generated by GenAI, in part or in whole, as your own (even in paraphrased form) constitutes an act of academic dishonesty; it is no different from asking another person to write your assignment or claiming others' ideas as yours.							
Student Study Effort Expected	Class contact:						
	▪ Lecture						39 Hrs
	Other student study effort:						
	▪ Assignment						40 Hrs
	▪ Self-study						40 Hrs
	Total student study effort						119 Hrs.
Reading List and References	1. Keegan, K. D. (2008). Biotechnology Valuation: An Introductory Guide. John Wiley & Sons Ltd. ISBN 9780470511787						
	2. Gruber, A. C. (2009). Biotech Funding Trends: Insights from Entrepreneurs and Investors. Publisher. ISBN: 3527324356						
	3. Rogers, J. (2019). *Strategy, value and risk: Industry dynamics and advanced financial management* (4th ed.). Palgrave Macmillan. ISBN : 9783030219772 (print)						